

17SLC1

SomeonecoolLovesMaths

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Problem

A rectangle $\mathcal{R}$ with odd integer side lengths is divided into small rectangles with integer side lengths. Prove that there is at least one among the small rectangles whose distances from the four sides of $\mathcal{R}$ are either all odd or all even.

Claim 1 — A small rectangle with odd sides always satisfies the conditions.

Let the coordinates of the original triangle be $(0,0), (a,0), (0,b), (a,b)$ such that $a \equiv b \equiv 1 \pmod{2}$. Consider a small rectangle inside the rectangle with coordinates $(p,q), (p,r), (s,q), (s,r)$. Say p is odd wolog. Then $s \equiv 0 \pmod{2}$ and $p - 0 \equiv 1 \pmod{2}$. So, $s - a \equiv 1 \pmod{2}$. Similarly we can proceed with q, r . $\square$

Claim 2 — There is a rectangle with odd area.

Say not. As a result all of the areas are even, so they must sum to an even number, which is equal to the area of the final rectangle which is odd, impossible! As a result there must exist a rectangle with all odd sides. $\square$